# Prabigya Acharya

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AI researcher, teacher and AI communicator. My research spans natural language processing for low-resource languages, LLM-based agentic systems and vision–language understanding of 3D scenes, with an emphasis on building benchmarks and evaluating systems together with the practitioners who use them. I lead an undergraduate research centre, teach AI from high-school workshops to professional training, and communicate the advantages, risks and responsible use of AI to different audiences.

## Education

**Bachelor's in Computer Engineering** 2019 – 2024
*IOE, Pulchowk Campus, Tribhuvan University • GPA: 81.02 / 100 (Distinction)*

- Ranked in the top 1% among ∼16,000 national applicants in the IOE entrance examination; ranked 14th of 96 in the graduating class.
- Received merit scholarships for 6 of 8 semesters for academic excellence.
- Final-year thesis: *3D Exhibition Environment using 3D Gaussian Splatting* — a 3D heritage reconstruction pipeline with an interactive exhibition environment. Supervised by Dr. Jyoti Tandukar (major project) and Prof. Dr. Subarna Shakya (minor project).
- Undergraduate research on lightweight language models for PII masking (with Liza Shrestha): fine-tuned T5-small and Mistral-7B, exceeding the published baseline on the ai4privacy dataset, and deployed the models live as a Discord bot to study accuracy/latency trade-offs on real conversational text. Published as an arXiv preprint.

## Research and Work Experience

**Curriculum-Grounded Nepali Question Generation Based on Bloom's Taxonomy** Jun 2026 – Jul 2026
*Sole author • Accepted at the NLP4PI Workshop, EMNLP 2026*

- First study of Bloom's-taxonomy-conditioned question generation for Nepali, grounded in national curriculum objectives for Grades 7 and 10.
- Had two government Nepali teachers annotate curriculum objectives with Bloom's levels; near-chance single-label agreement led to a redesign as multi-label annotation.
- Benchmarked zero-shot LLMs and fine-tuned classifiers for Bloom's-level extraction, identifying systematic suppression of higher-order categories by LLMs.
- Built a bilingual annotation app for blind teacher evaluation of generated questions; taxonomy-guided generation raised teacher endorsement from 12.5% to 88%.
- Deployed the question-generation tool at a government school, where it is in use.

**Undergraduate Research Coordinator** Apr 2026 – Present
*Research & Incubation Center (RAIN), Sunway College Kathmandu • Kathmandu, Nepal (On-site)*

- Lead the college's research division; founded research groups for NLP and computer vision and coordinate ∼20 student researchers.
- Supervised four undergraduate research papers from conception to camera-ready submission; all four were accepted and presented at the 2nd MVIC National Research Conference, Nepal.
- Currently supervise five student projects: Nepali-language email readiness of `.np` mail servers; misinformation spread during the August 2026 floods; disaster-location extraction from social media (presented at ICIMOD); social-media sentiment during the Nepal GenZ movement; in-place 3D scene editing in Gaussian Splats; transfer of safety practices from VR chemistry classrooms.
- Principal investigator on *Auditing Nepal's Readiness for Nepali-Language Email*, the first systematic audit of internationalised email readiness across the `.np` domain; wrote the proposal that won NPR 200,000 in competitive funding from NPIX.
- Built the audit toolchain and ran a pilot that found most primary `.np` mail servers are unreachable from outside Nepal, motivating a dual-vantage design with a probe node inside NPIX.
- Founded and chair the Sunway Research Ethics Committee, the college's first ethics body overseeing research involving human participants and data.
- Advise a four-student team shortlisted among 60 teams worldwide in the Khalifa University–UNESCO Global Water Hackathon 2026.
- Lead outreach programmes, including an AI and BI workshop series for ∼200 high-school students.

**AI Engineer** Jun 2026 – Present
*Cyberensic • Australia (Remote, part-time)*

- Sole AI developer for a compliance platform; designed its retrieval-augmented architecture (hybrid dense and lexical retrieval with reranking).
- Built LLM-based gap assessment against more than ten standards, including ISO 27001, ISO 42001 and SOC 2, with human-review gates and evaluation of retrieval quality.

**Research Assistant** Jun 2025 – Mar 2026
*NAAMII (Nepal Applied Mathematics and Informatics Institute for Research) • Supervisor: Dr. Danda Pani Paudel*

- Designed and built the agentic pipeline for *SceneBench* that converts 3D Gaussian Splatting scenes into hierarchical descriptions (scene → room → functional area → object group → object).
- Built the human-in-the-loop annotation pipeline behind a benchmark of 969 photorealistic scenes and 33.5K questions on existence, spatial reasoning and multi-step reasoning.
- Tools: Python, PyTorch, LangGraph, Gemini API, 3D Gaussian Splatting.

**Research Intern** Nov 2024 – Apr 2025
*NAAMII • Supervisor: Dr. Ajad Chhatkuli*

- Reproduced the closed-source inference loop of GenWarp (NeurIPS 2024) for single-image novel-view synthesis with Stable Diffusion.
- Built a pipeline to extract frames from RealEstate10K videos and derive depth and normal maps, curating a paired image–depth–normal dataset.

**Software Engineer** Apr 2024 – Nov 2024
*Growthzilla • Remote*

- Designed and implemented an online-payment integration and an automated reconciliation and reporting system for the Uzeli POS platform.
- Operated independently in a distributed team across three countries.

## Publications

1. **Acharya, P.** (2026). *Generating Curriculum Grounded Nepali Questions Based on Bloom's Taxonomy.* Accepted at the NLP for Positive Impact Workshop (NLP4PI), EMNLP 2026.
2. Khanal, A., **Acharya, P.**, Poudel, R., Kapali, S., Bhatta, B., Paudel, P., Rameau, F., Paudel, D. P. (2026). *SceneBench: A Hierarchical Benchmark for Vision-Language Understanding of 3D Scenes.* Under review at the International Conference on 3D Vision (3DV) 2027.
3. **Acharya, P.**, Shrestha, L. (2025). *A Comparative Study of Light-weight Language Models for PII Masking and their Deployment for Real Conversational Texts.* arXiv:2512.18608.
4. Nwana, N., Dahal, C., **Acharya, P.** (2026). *Basketball IQ vs. Kalshi: Evaluating Prediction Market Accuracy Against Statistical Models in College Basketball Forecasting.* Manuscript complete; journal under selection.
5. Poudel, R., **Acharya, P.** (2026). *To What Extent Do Large Language Models Exhibit Behavioural Patterns that Correlate with Human Decision-Making in Recursive Strategic Prediction Scenarios?* 2nd MVIC National Research Conference, Nepal.
6. Neupane, M., Luitel, S., Poudel, R., **Acharya, P.** (2026). *Convenience for Whom? A Two-Sided Study of Ride-Hailing in Nepal.* 2nd MVIC National Research Conference, Nepal.
7. Regmi, R., **Acharya, P.** (2026). *Nigrani: A Multi-Framework Machine Learning System for Fraud Detection.* 2nd MVIC National Research Conference, Nepal.
8. Adhikari, P., **Acharya, P.** (2026). *Towards Fair Fares: Designing a Resilient Contactless Payment Ecosystem for Public Transit in Nepal.* 2nd MVIC National Research Conference, Nepal.

## Grants and Awards

**NPIX Research Grant** 2026
*Internet Exchange Nepal (NPIX) • NPR 200,000 • Principal Investigator*

- Competitive grant, awarded in full, for *Auditing Nepal's Readiness for Nepali-Language Email*, a four-month national study of internationalised email readiness across the `.np` domain.
- Proposed, budgeted and lead the project; NPIX provides a probe node inside its network and the full `.np` zone data.

**Shortlisted, Khalifa University–UNESCO Global Water Hackathon** 2026
*Advisor to the Sunway College team • Top 60 teams worldwide*

- Guided the team's proposal on water reuse and circularity through a highly competitive global selection to the final round.

**Merit Scholarship** 2019 – 2024
*IOE, Pulchowk Campus, Tribhuvan University*

- Semester fee waiver awarded on academic standing for 6 of 8 semesters of the Computer Engineering programme.

## Technical Skills

- **Programming & Frameworks:** Python, PyTorch, TensorFlow, Hugging Face, LangChain, LangGraph, FastAPI, Flask, Node.js, React
- **Machine Learning & AI:** LLMs and agentic systems, RAG and retrieval evaluation, LoRA/QLoRA fine-tuning, transformers, computer vision, 3D Gaussian Splatting, benchmark and dataset design, human-in-the-loop evaluation
- **Data & Tools:** PostgreSQL/pgvector, statistical testing (Cohen's $\kappa$, Kruskal–Wallis), Three.js, Blender, OpenCV, Git, LaTeX

## Certifications and Programs

- **Engineering License**, Nepal Engineering Council — licensed computer engineer, registration number 87259 (2024).
- **4th NAAMII AI School** (2024) — intensive 10-day AI school (55 h lectures, 18 h labs, 14 h projects) taught by experts from ETH Zürich, NUS and other institutions; covered ML, NLP, transformers, GNNs and responsible AI.

## Teaching, Talks and Outreach

**Panellist and Speaker** Oct 2026
*5th National Conference on Disaster Risk Reduction, ICIMOD, Lalitpur*

- Presented *Research-driven Student-Led Innovations for Disaster Management* — five RAIN-supervised student prototypes — in the session on data, technology and innovation in disaster risk management.

**AI Trainer for the Legal Profession** May 2026 – Oct 2026
*Nepal Bar Council, Nepal Bar Association, Special Court Bar Association*

- Designed and solely delivered the "Legal Profession and AI" training series; ∼1,200 legal practitioners trained across five events in Kathmandu, Lalitpur and Butwal.
- Delivered the graduation address *AI Ethics and Usage in the Legal Profession* at the Nepal Bar Council before 800 newly licensed advocates.

**Facilitator** 2026
*AI Safety Nepal fellowship (AI Safety Atlas curriculum)*

- Facilitated a weekly discussion group of 6–8 participants across Nepal's first dedicated AI safety fellowship.

**Teaching Assistant** 2025
*6th Nepal AI School, NAAMII*

- Supported lectures, workshops and project sessions of a 10-day international AI school (100+ participants) led by researchers from Oxford, ETH Zürich, IIT and other institutions.

**Instructor, Software Fellowship** 2023
*LOCUS, Pulchowk Campus*

- Taught data structures and algorithms in Python to a class of 50 first-year students for 4 of the 7 fellowship days.

**Volunteer, Children in Technology** 2023
*LOCUS, Pulchowk Campus*

- Delivered introductory computer literacy workshops to 40+ children from underserved communities.